

Grammar Gus: Claudia's Audit and Writing Scaffold Plan

For: the teacher of the 5th and 6th grade machine builders **Date:** 2026-10-07

Teacher's request: "run until this dev plan is complete for grammar gus, then claudia should evaluate and audit all features for design, function, gamification, and accuracy for academics. claudia should then take that report, evaluate what features the platform doesn't have (grammar gus), what it could have, and research based on academic components that need to be instructed for writing and grammar, and scaffold them in a fun gamified way for my science loving, machine building kids"

This report has three parts:

1. **The audit** of everything on Gus's Workboard, and what was fixed today.
 2. **The gap analysis and research base:** what Gus does not teach yet, and why the next features are built the way they are.
 3. **The scaffolded, gamified machines** to build next, with a build order.
-

Part 1. The audit

Claudia read the code, not just the screens, and checked four areas. The dev plan run was finished first (rounds 8, 9 and 10), so the audit covers the whole Workboard.

Design

What works:

- A play-first sandbox with one clear "Next" step.
- Symbol-shaped machines in their symbol colors.
- A docked checklist.
- Sound and motion on every part, with a Calm mode.

What she flagged:

1. **Overlaps with the machine.** The "Capital letter?" helper sat on top of the sentence plate, the COMBO banner covered parts snapped on top, and long job banners ran off the board. **Fixed.**
2. **Snapped-on parts too small to tap** (a comma could be about 34px on an iPad). **Fixed:** every snapped-on part now has an invisible 44px-plus tap area at any zoom.
3. **Calm mode was two taps deep and forgotten each time.** **Fixed:** it is back as a 🌙 Calm button on the top bar, and the iPad remembers it.

4. **Low contrast and flashing. Fixed:** darker Spare Parts Bin and move grip; the Dead-End Detector light no longer blinks; the duplicator screen no longer flickers.
5. **Copy rule.** The Time Warp Toaster said "past · now · next" and two Gus lines said "now". **Fixed:** they now say "past, present, future", "Pull the Start Lever!" and "at the moment".
6. **Still open (queued):**
 - Too much on screen at once for some students.
 - About 40 fun parts in two long lists.
 - Some hard words in the everyday word lists (zealously, upon).
 - A few tiny labels inside the drawings.

Function

Fixed today:

- **Tapping a comma** used to drop it at the end, which is always wrong. It now snaps on top of the word before the joining word. If there is no joining word, Gus explains where a comma goes.
- **Snapping a part onto a taken spot** used to delete the old part silently. The old part now goes into the Spare Parts Bin and Gus says so.
- **Gear farming:** pulling the lever again on the same sentence kept paying gears. Gears now pay once per sentence.
- **Combos** were only "are the parts there?". Plural Pileup now needs a real plural noun, and Name Parade needs a name in a list of three.
- **Gus's review** said "Describe the mia" and "What did the he do?". Names now get capital letters, names and pronouns get no "the", and the quick question says "What does...", "What did..." or "What will..." to match the time.
- **Time words lowercased names** ("Then, mia ran."). Fixed.
- **Review tips suggested parts that cannot follow an object yet.** They now appear only when they will work.
- **Pronoun Teleporter and Cannon with a team of two** ("the cat and the dog"): they now use "they". The Cannon also uses him, her or them after the action.
- **Read-aloud now honors the sound switch.**
- **Teacher reports:** a missing TV, capital letter part or punctuation part is no longer logged as a failed grammar attempt.

Still open (queued):

- Undo does not bring back a part put in the Spare Parts Bin.
- The board saves on the iPad, not to the teacher's account.

Gamification

What is right:

- Choice everywhere.
- Rewards for quality, never for speed.
- Mistakes stay private.
- No leaderboards.
- No gambling: combos are caused by the grammar, and the word shuffler pays nothing.

Fixed: the Slot Machine is now the **Word Shuffler** (" Shuffle!", a neutral sound and no "KA-CHING") so it never feels like a casino.

Fixed: gadgets no longer silently fix a student's mistakes at every level. This is the **fade ladder**:

Help level	What a gadget does
Full	Fixes the mistake and says the rule.
Guided	Lights up the machine with the mistake and names the rule. The student fixes it.
Challenge	Says only "one thing is off". The student finds it.

It applies to the A/An Sniffer, Agreement Gears, Describe Sorter, Pronoun Turnstile, Pronoun Teleporter and Pronoun Cannon. Recalling a rule teaches more than being handed the answer.

Still open: gears have nothing to spend on (see the Paint Shop in Part 3), and the Workboard has no teacher-shared showcase yet.

Academic accuracy

Checked and correct:

- a and an by sound
- plural possessives (the cats')
- adjective order
- questions with do, does, did and will, and short answers
- quotation marks (comma inside, said and asked)
- he and him, I and me
- the opener comma
- past, present and future forms

Fixed today: each of these taught a wrong or incomplete rule before.

What was wrong	The correct rule now taught	Standard
No comma needed before "and" joining two whole ideas	A comma goes before and, but, or, so and yet when they join two whole ideas: "The cat runs, and the dog jumps."	L.4.2.c

What was wrong	The correct rule now taught	Standard
The Comma Drawbridge passed "The cat, and the dog run"	An extra comma before a team of two is caught	L.5.2
"so" was taught as a "because" word	"so" joins two whole ideas with a comma. The Because Seesaw now holds because, when, after, before and while	L.4.2.c, L.3.1.h
"for" between two ideas	Removed (old-fashioned, and it collides with the preposition). "so" and "yet" added	L.3.1.h
Extra commas and capital letters were never caught	"No extra commas" and "Capital letters only where they belong" are checklist rows, and the machine will not run with them	L.5.2, L.1.2.a
"a tallest cat"	Superlatives take "the": the tallest cat	L.3.1.g
"more red", "purpler"	redder and reddest; purple and orange use more and most; striped, wooden and cosmic do not come in sizes	L.3.1.g
"wolves", "potatos", "the cat haves"	wolves, leaves, knives, potatoes, heroes; has	L.3.1.b, L.3.1.d
"Does quickly, the cat jump?"	The Question Crane will not lift a sentence with an opener	L.1.1.j

Still open:

- Reflexive pronouns are not checked against the who ("The dog saw themselves").
- "they" for one person of unknown gender. Some tests mark it as a mismatch; the teacher may want to choose.
- A gentle comma after a mild interjection ("Oops, I spilled").
- Pronouns after a preposition ("with her"), which cannot be built yet.
- A "because" idea at the front of the sentence.

Part 2. What Gus does not teach yet, and the research behind the next steps

Gaps in sentence grammar and conventions (L.3 to L.6)

- **Linking verbs** (is, am, are, was, were). Every verb machine today is an action, so "The magnet is strong." cannot be built.
- **A joining idea first**, with its comma: "When the gear spins, the lamp glows."
- **Fragments and run-ons**: no machine shows what is wrong with them.
- **Sentence combining and appositives**: "Gus, a robot, fixed it."
- **Progressive and perfect tenses** (is spinning, has spun) and **modal verbs** (can, must, might).
- **Tense consistency across a whole paragraph**.

- **Relative clauses** (who, which, that).
- **Where words and how words after an object:** "The cat chased the ball under the table."
- **Pronouns after where words:** "with her", not "with she".
- **Interjection commas, titles, and dialogue paragraphs** (a new speaker starts a new line).
- **Homophones and commonly confused words** (their, there, they're; its, it's; than, then), and **word parts** (prefixes, suffixes, roots).

Gaps in writing craft (W.3 to W.6)

- Sentence expansion that answers who, what, when, where, why and how.
- Varied openers, and transitions between paragraphs.
- Topic sentence, supporting details and a concluding sentence. Blueprints give a shape, but the parts are not named or checked.
- Informative writing with facts and science words.
- Opinion writing with reasons and evidence.
- Revising your own writing.
- Choosing the stronger, more precise word.

Gaps in science writing language

- **Procedures:** numbered steps, time-order words and command verbs.
- **Cause and effect, and compare and contrast** (because, so, however, both, unlike).
- **A hypothesis** ("If... then...") and **observations** in the past.
- **Claim, Evidence, Reasoning (CER).**
- **Labels on diagrams, and quantities with units** (12 cm, 3 seconds).

The research base

- **Sentence combining** (Saddler and Graham; Graham and Perin, *Writing Next*): practice joining short sentences into longer ones improves sentence quality and overall writing.
- **The Writing Revolution** (Hochman and Wexler): sentence-level work taught explicitly: because, but, so; appositives; expanding a kernel sentence with who, what, when, where, why; joining words.
- **SRSD, Self-Regulated Strategy Development** (Harris and Graham): explicit strategy steps, modeling and mnemonics, with support faded over time. It has strong evidence for students with learning disabilities and ADHD.
- **Grammar taught in context** (Myhill and colleagues): grammar tied to the student's own writing and choices helps writing. Isolated drills transfer poorly. This is why every fun part has a grammar job.

- **Worked examples and fading** (Sweller; Renkl and Atkinson): a full example first, then a partial one, then none. This is the fade ladder and "I do, we do, you do".
- **Retrieval practice** (Roediger and Karpicke): recalling a rule beats being handed it. This is why gadgets stop fixing mistakes at Guided and Challenge.
- **Morphology instruction for dyslexia** (Bowers, Kirby and Deacon): teaching prefixes, suffixes and roots helps reading and spelling.
- **Claim, Evidence, Reasoning** (McNeill and Krajcik): a three-part scaffold that improves students' scientific explanations.
- **Self-Determination Theory** (Deci and Ryan): autonomy (choosing what to build), competence (visible growth) and relatedness (a teacher-shared showcase).
- **Universal Design for Learning** (CAST): picture, sound and text for everything, and more than one way to answer.
- **No leaderboards**: ranking is the most-flagged risk for neurodivergent learners. Every feature is personal progress only.

Part 3. Scaffolded, gamified machines to build next

Every machine has sound, motion and a Pixel TV payoff, and none of them is a gamble. Each one uses the help levels: Full, then Guided, then Challenge.

Build tiers:

- **Tier A** fits the current engine.
- **Tier B** needs a small engine addition.
- **Tier C** is later.

#	Machine	What kids do	Skill (standard)	Scaffold	Tier
1	Logic Gate	Two sentence halves meet a gate: BECAUSE (cause), BUT (surprise) or SO (result). The gate swings and lights a different pipe. A Flip Switch moves a "because" or "when" idea to the front, and the comma has to follow it.	Joining words, the comma before but and so, a joining idea first (L.3.1, L.5.2, W.4.2)	Gus builds one; the student picks the gate and Gus places the comma; then the student does both	A
2	Fusion Reactor	Drop in 2 or 3 short sentences ("The robot is small." "The robot is fast."). Repeated words are crushed and one smooth sentence rolls out ("The small, fast robot...").	Sentence combining (L.4.1, L.5.1, W.5.5)	Full: repeats removed for them. Guided: repeats glow. Challenge: they choose what to cut	B

#	Machine	What kids do	Skill (standard)	Scaffold	Tier
3	Appositive Clamp	Clamp a fact about a noun into a sentence ("Mr. Cruz, our science teacher, fixed the lab."). Two comma tabs snap on either side.	Appositives and their commas (L.5.2, W.5.5)	Full: commas snap on. Guided: gaps glow. Challenge: runs only with both commas	B
4	Spark Check: Fragment and Run-on Detector	Sentences ride a belt through a scanner that asks "Who or what? What happens?" A missing part is a dud; two ideas glued together is a jam. Then fix it.	Complete sentences, fragments, run-ons (L.3.1, L.5.1, L.6.1)	One clearly broken sentence first, then more varied ones. Icons and sounds, never color alone	A (fragments), B (run-ons)
5	Equals Sign Machine	A balance scale: who = what it is like ("The magnet is strong."). Action verbs spin a motor; is, am, are, was and were show an equals sign.	Linking verbs (L.4.1, L.5.1)	Full: Gus picks is or are. Guided: gears show the mismatch. Challenge: sort mixed verbs into motor or equals	A
6	CER Lab Report	Three linked machines: the Claim Cannon, the Evidence Conveyor (a data card with a number and unit), and the Reasoning Bridge ("This shows that... because..."). It runs only when the bridge connects both.	Scientific explanation, reasons and evidence (W.6.1, W.5.7, NGSS Practice 7)	SRSD style: Gus models a full report, then sentence starters, then blank machines	B
7	Hypothesis Engine	Pick a variable ("If I raise the ramp...") and a prediction ("...then the marble will roll faster."). A test chamber runs it. Then write the observation in the past.	If... then sentences, future and past (W.5.7, L.5.1)	Full: frame fixed. Guided: choose both parts. Challenge: write it all	A
8	Procedure Conveyor	Numbered steps ride a belt ("First, pour the water."). Put scrambled steps in order. Each step needs a command verb and a time-order word.	Command verbs, sequence words, procedures (W.3.3, W.4.2)	Full: ordered. Guided: the wrong step wobbles. Challenge: write the missing step	A
9	Transition Track	Sentence cars couple with a transition (also, however, as a result, for example). Each coupling is a type: adds, contrasts, shows a result, gives an example. The train only moves with the right one.	Transitions (W.4.2, W.5.2)	2 choices, then 4 with type icons, then the full set	A

#	Machine	What kids do	Skill (standard)	Scaffold	Tier
10	Revision Workshop (ARMS)	Load a rough paragraph. Four tools: Add (a wrench adds detail), Remove (a cutter trims repeats), Move (a crane moves a sentence), Substitute (a swap arm trades a dull word for a precise one). The TV replays after each repair.	Revising and editing, precise words, tense across a paragraph (W.4.5, W.5.5, L.5.3)	Full: one tool per pass. Guided: Gus points to a weak spot. Challenge: the student chooses	B
11	Homophone Sorter	their, there and they're drop into three chutes. A sentence clue and a picture light the right bin.	Commonly confused words (L.3.2, L.4.2)	Pairs first, then trios; the picture hint fades	A
12	Gus's Paint Shop (gear sink)	Spend gears on machine paint, decals and new factory floors (a lab, a space station, a junkyard). Gears never buy hints, help or retries.	Autonomy and renewed novelty (SDT)	Never gates help or Calm mode	B
13	Showcase Wall	Send one finished machine to the teacher. Approved ones appear on a class wall with a sticker. No counts, no rankings, no comments, and mistakes are never shown.	Writing for an audience (W.3.4), relatedness	Teacher-approved only	C
14	Expansion Rig	A short sentence ("Gus builds.") with six ports: who, what, when, where, why, how. Plugging a part in adds a phrase after the action or object ("with a wrench", "in the lab", "with her"), and the TV scene grows.	Sentence expansion, where phrases after an object, pronouns after where words (L.4.1, L.5.1, W.3.3)	Full: ports labeled with a picture and word. Guided: only the question shows. Challenge: Gus asks, the student chooses the port	A
15	Label and Unit Maker	Drag labels onto a diagram of a machine the student built, and write measurements with a number and a unit ("12 cm"). A wrong unit sparks. Word parts show in colors with labels.	Labels, units, science words, morphology (L.4.6, W.5.2)	Word bank, then picture hints, then none	B
16	Compare and Contrast Balance	"both" and "unlike" words weigh each side of a scale.	Compare and contrast (W.4.2, W.5.2)	Later	C

The fade ladder, everywhere

The same gadget changes with the help level, so the student recalls the rule instead of being rescued. It is now live on six gadgets. Next it should reach the Capital Letter Press, the punctuation parts and the

Comma List Train. Each student's level could differ by gadget, set by the teacher. A student who succeeds twice in a row can be offered the next level by choice, never forced. Failure stays private.

Recommended build order

Phase 1 (uses the current engine; closes the biggest gaps):

- the fade ladder on the remaining parts
- Logic Gate
- Equals Sign Machine
- Expansion Rig
- Spark Check (fragments)
- Homophone Sorter
- Hypothesis Engine, Procedure Conveyor and Transition Track (science writing first)

Phase 2 (small engine additions):

- run-ons and the fix mini-game
- Appositive Clamp and Fusion Reactor
- Revision Workshop
- CER Lab Report
- Label and Unit Maker
- Gus's Paint Shop

Phase 3 (later):

- Showcase Wall (needs a teacher approval flow)
- relative clauses, progressive and perfect tenses, modals, titles, dialogue paragraphs
- Compare and Contrast Balance and opinion-writing machines
- seasonal Paint Shop floors

Which audit gap each one closes:

Audit gap	Closed by
Comma before a joining word and a joining idea first	Logic Gate
Linking verbs	Equals Sign Machine
Where phrases after an object, and pronouns after where words	Expansion Rig
Extra commas, capital letters and tense across a paragraph	Revision Workshop
Gears with nothing to spend on	Paint Shop
No teacher-shared showcase	Showcase Wall

Every new machine needs the iPad check before sign-off:

- a tap path for every drag
- 44px targets with spacing
- portrait and landscape layouts